

QArchGauge procedure

same input, native deadline, quantum circuit metadata, and projection knobs stay in one case record

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1. Shared application case: input, seed, quality metric, tolerance

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2. Native HPC path
 T_{native}, Q_{native}
exact / Krylov / heur.

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3. Quantum-circuit path
 T_{qsim}, Q_q
VQE / QAOA / Trotter

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4. Auditable case record
runtime, quality gap,
gates, shots, evals

Logical ops
 t_1, t_2, t_m

Shot lanes
 P_{shots}

Decode/control
 T_{err}

Quality recovery
 R_q

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Runtime gate
 $T_{qhw} < T_{native}$

Target map
amount + location